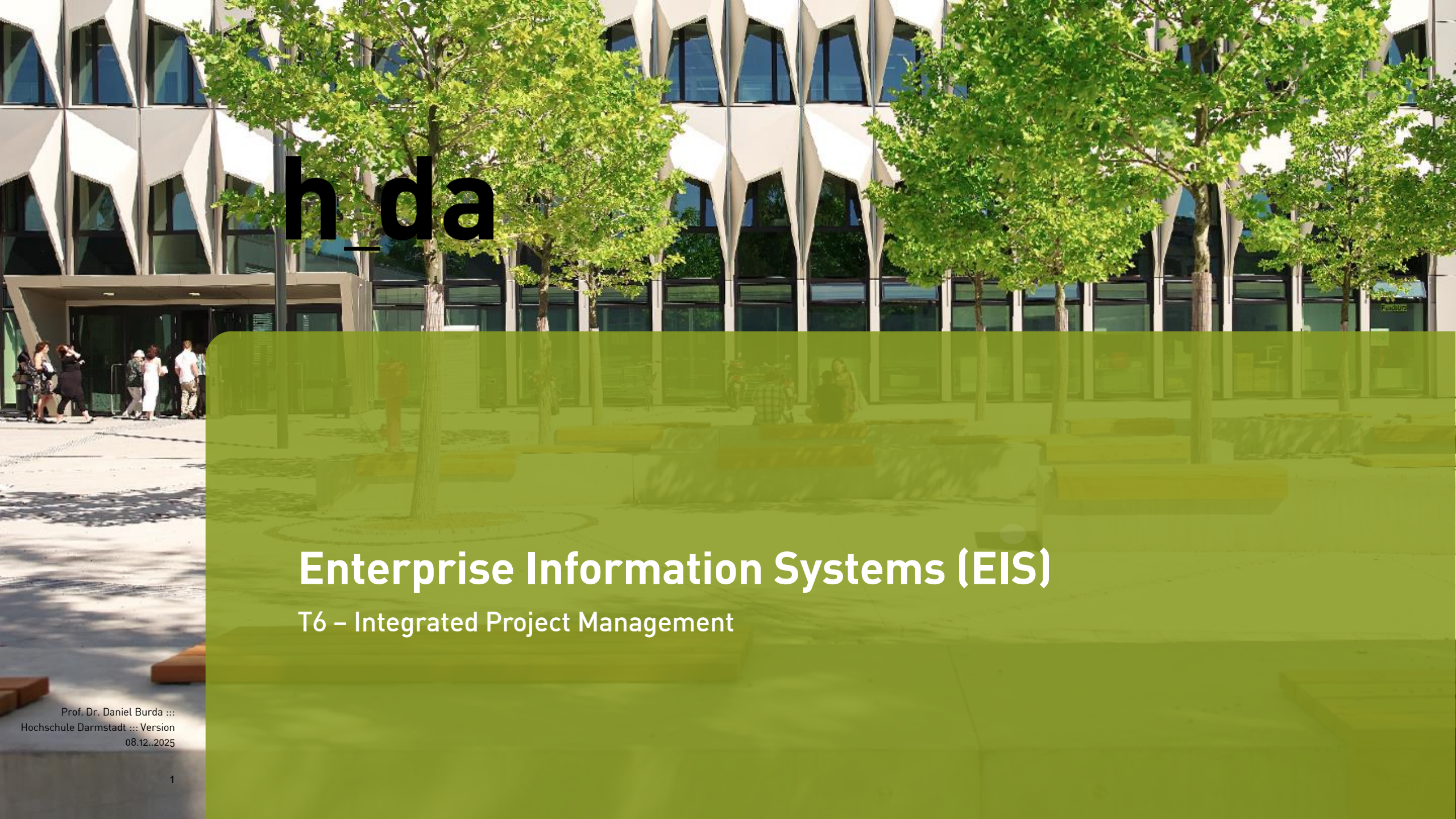


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Enterprise Information Systems (EIS)

T6 – Integrated Project Management

Prof. Dr. Daniel Burda :::
Hochschule Darmstadt ::: Version
08.12..2025

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Schedule (When?)

Lecture				Practical			
No.	Date/Time/Room	Topic No.	Topic	PVL Relevant	Topic	Practical No.	Dates (2 Groups)
1	Tue., 07.10.2025, 10:15 - 11:45 D14 / 04.04	T0	Intoduction & Module Overview	---	---	---	---
2	Tue., 14.10.2025, 10:15 - 11:45 D14 / 04.04	T1	Enterprise Information Systems - Theoretical Perspective	---	---	---	---
3	Tue., 21.10.2025, 10:15 - 11:45 D14 / 04.04	T2	SAP Overview and Introduction S/4 HANA	---			
4	Tue., 28.10.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (1)	---	SAP Navigation Exercises	P#1	28.10.2025 / 29.10.2025
5	Tue., 04.11.2025, 10:15 - 11:45 D14 / 04.04	T3	Sales and Distribution Processes (2)	x	SD Case Study	P#2	04.11.2025 / 05.11.2025
6	Tue., 11.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (1)	x	SD Case Study	P#2	11.11.2025 / 12.11.2025
7	Tue., 18.11.2025, 10:15 - 11:45 D14 / 04.04	T4	Materials Management and Procurement Processes (2)	x	MM Case Study	P#3	18.11.2025 / 19.11.2025
8	Tue., 25.11.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (1)	x	MM Case Study	P#3	25.11.2025 / 26.11.2025
9	Tue., 02.12.2025, 10:15 - 11:45 D14 / 04.04	T5	Financial Accounting Processes (2)	x	FI Case Study	P#4	02.12.2025 / 03.12.2025
10	Tue., 09.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (1)	x	FI Case Study	P#4	09.12.2025 / 10.12.2025
11	Tue., 16.12.2025, 10:15 - 11:45 D14 / 04.04	T6	Integrated Project Management (2)	x	PS Case Study	P#5	16.12.2025 / 17.12.2025
12	Tue., 13.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (1)	x	PS Case Study	P#5	13.01.2026 / 14.01.2026
13	Tue., 20.01.2026, 10:15 - 11:45 D14 / 04.04	T7	Human Capital Management Processes (2)	x	HCM Case Study	P#6	20.01.2026 / 21.01.2026
14	Tue., 27.01.2026, 10:15 - 11:45 D14 / 04.04	n/a	Test Exam / Q&A Final Exam	x	HCM Case Study	P#6	27.01.2026 / 28.01.2026

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Learning Objectives (What?)

Students can…

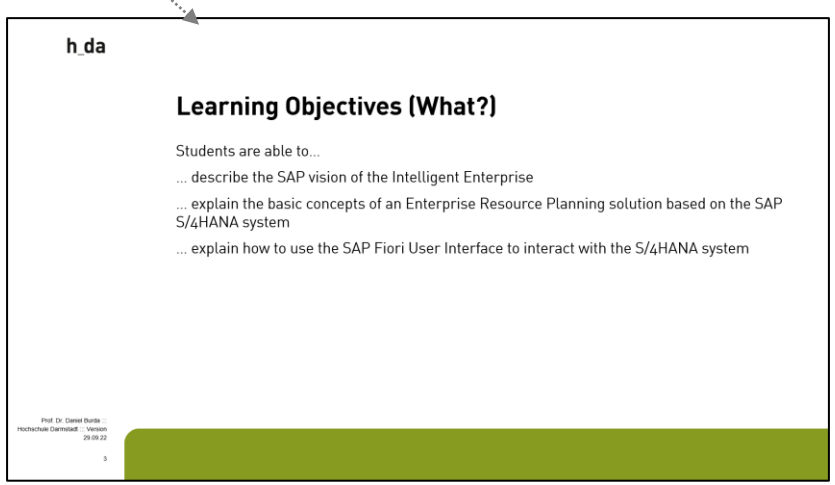
- explain why organizations employ projects to achieve their objectives.
- explain the relationships between portfolios, programmes and projects.
- name important processes in Project Management.
- describe a generalized project planning approach including major planning artefacts.
- explain the concepts of a work breakdown structure (WBS) and network chart.
- explain relevant master data important for the PS module.

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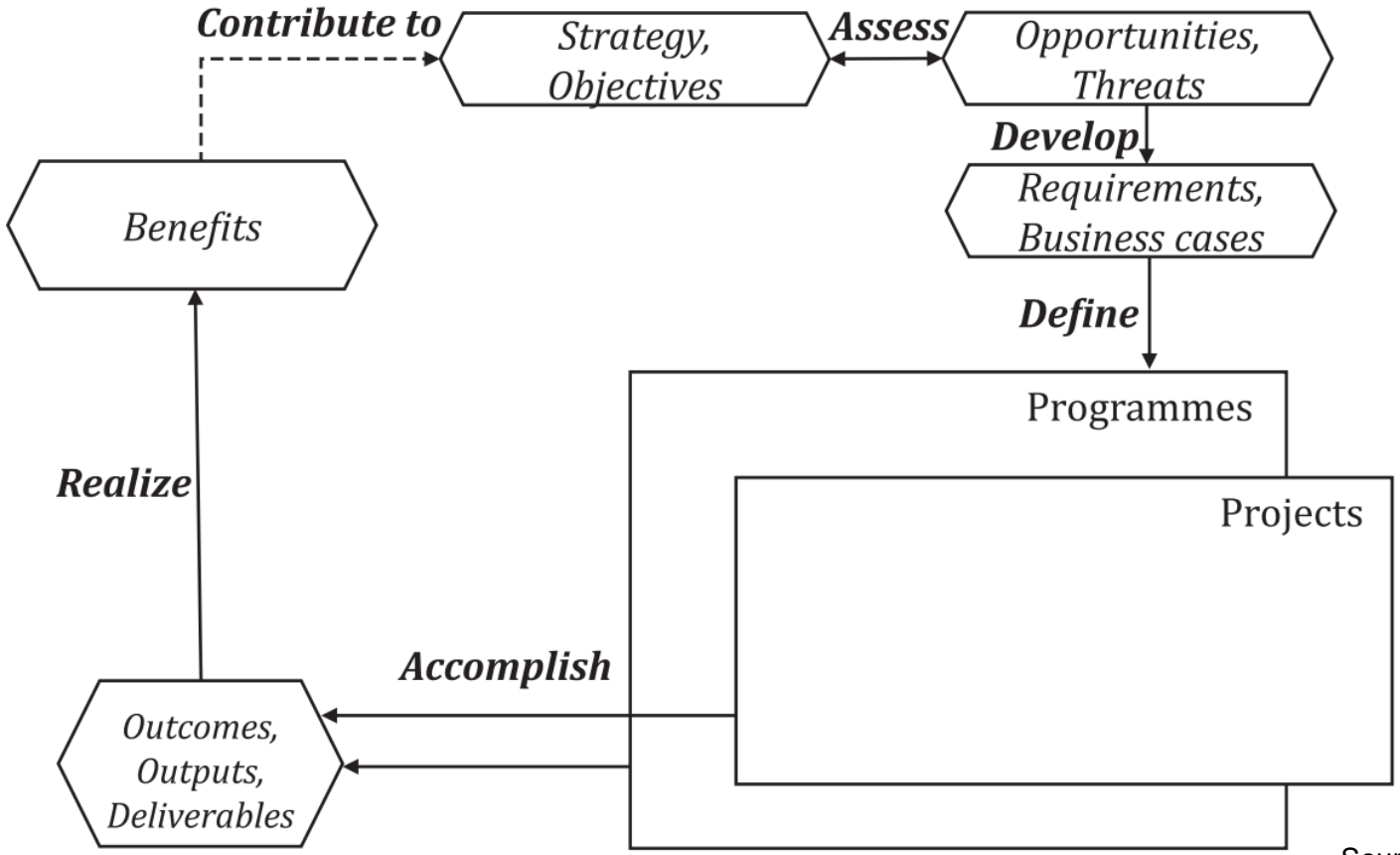
Agenda

- General Foundations Project Management
- PS Organizational Structure
- PS Master Data
- PS Processes
  - Project Planning
  - Project Execution
  - Project Closure
- Outlook PS Case Study



General Foundations: Project Management

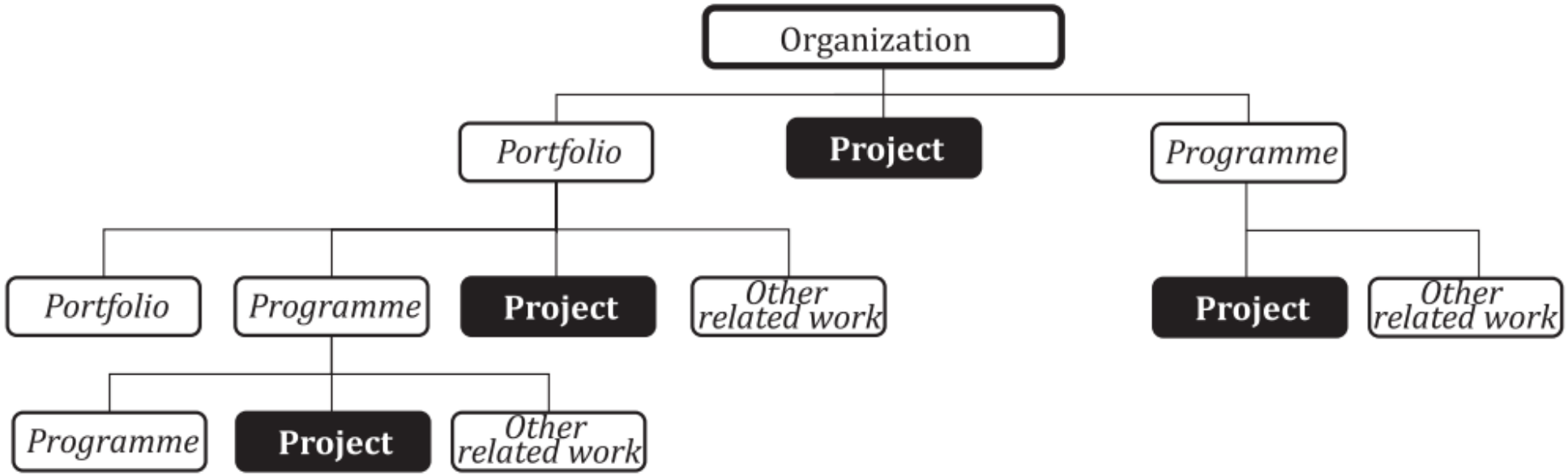
Projects and Programmes create value for organizations



Source: E DIN ISO 21502:2021-12, p. 7

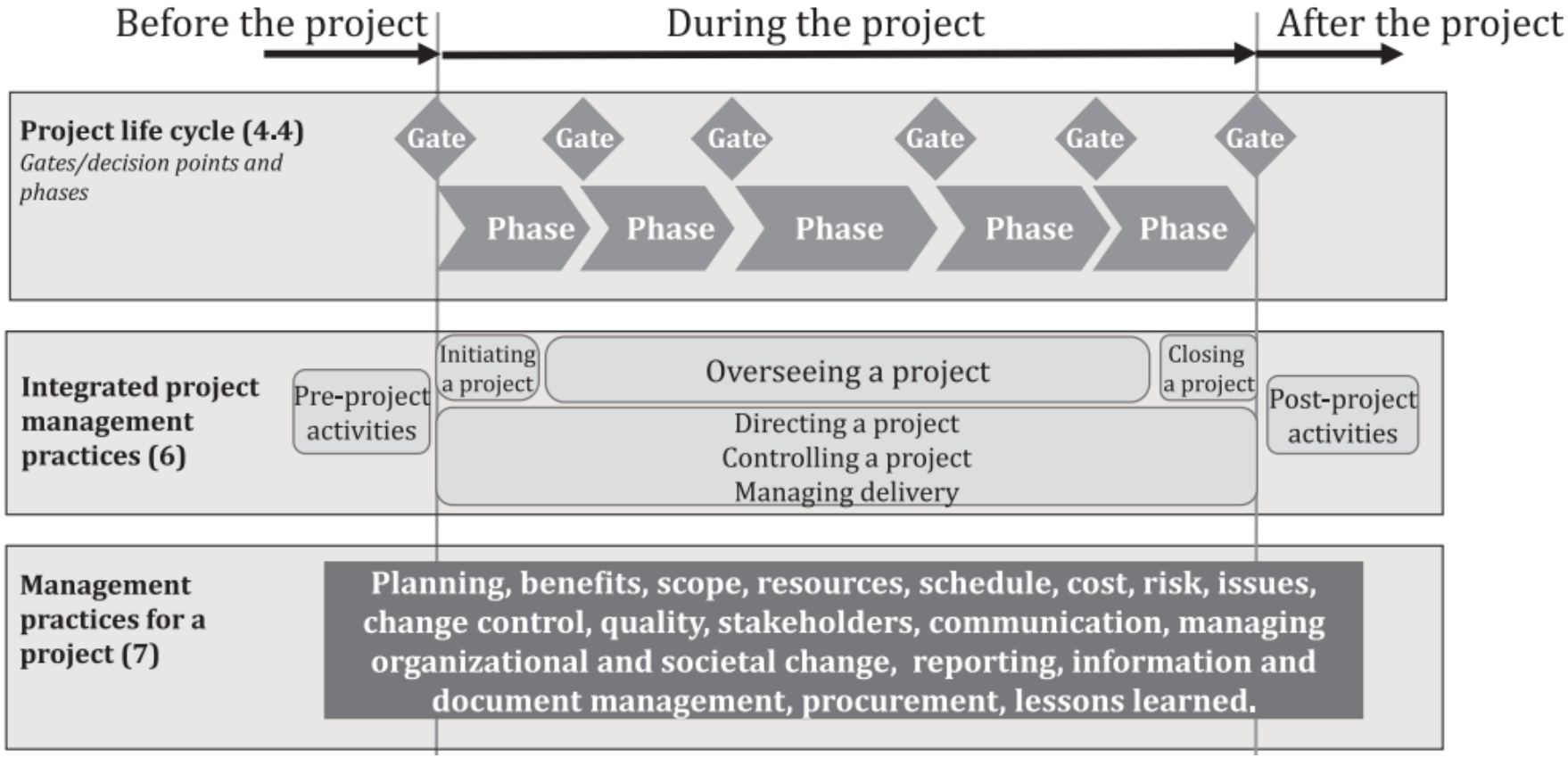
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Relationships between portfolios, programmes and projects



Source: E DIN ISO 21502:2021-12, p. 8

Relationship between project life cycle, integrated project management practices and management practices



Source: E DIN ISO 21502:2021-12, p. 10

Project Management Process (Groups)

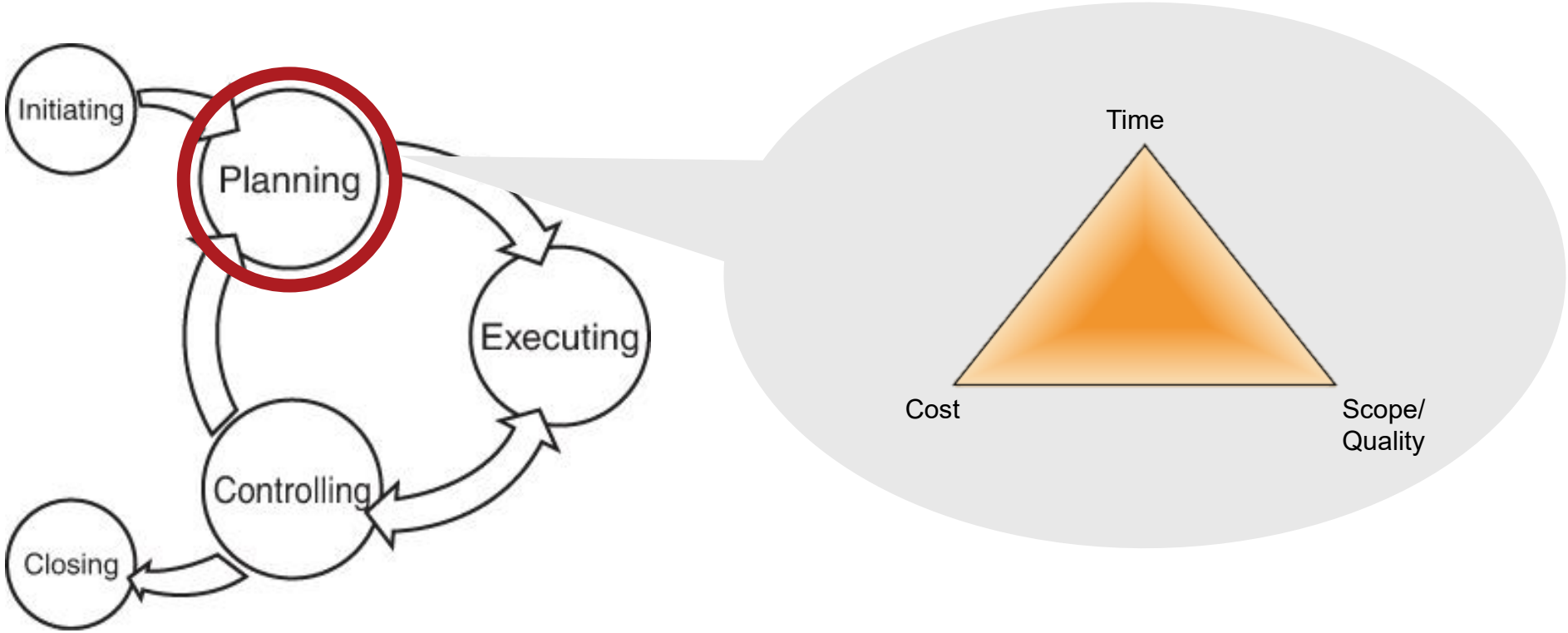
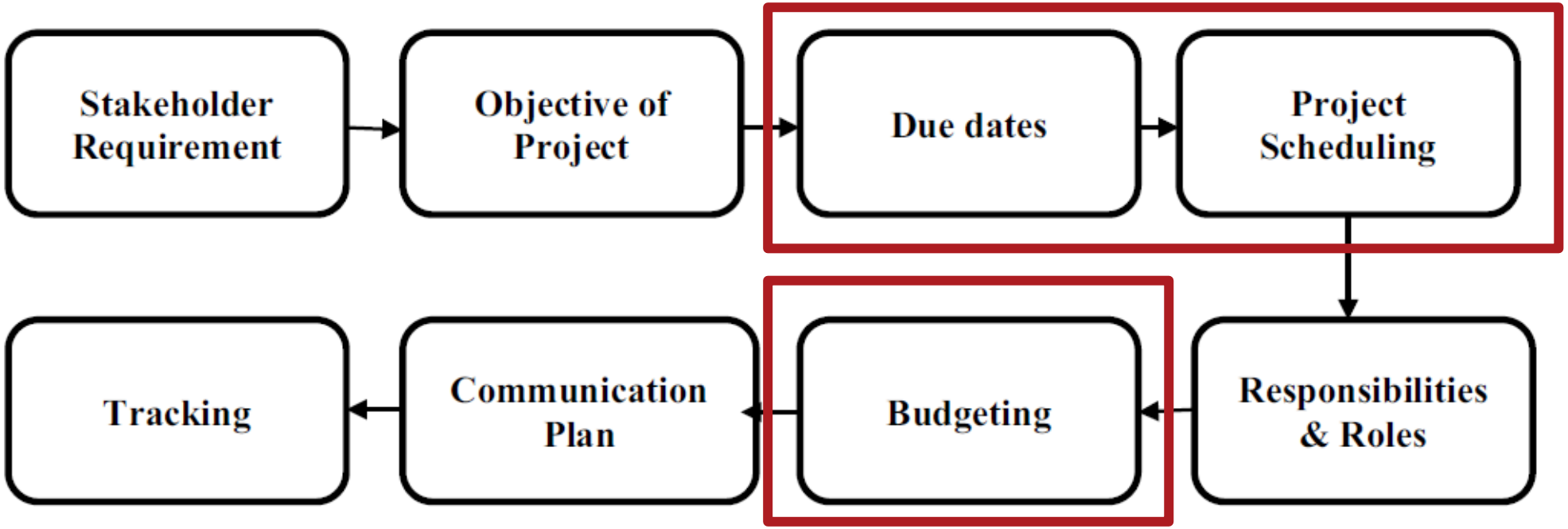


Figure 5.1, Horine, G. 2022. *Project Management Absolute Beginner's Guide*, (5th ed.). Sydney: Pearson Education, Limited.

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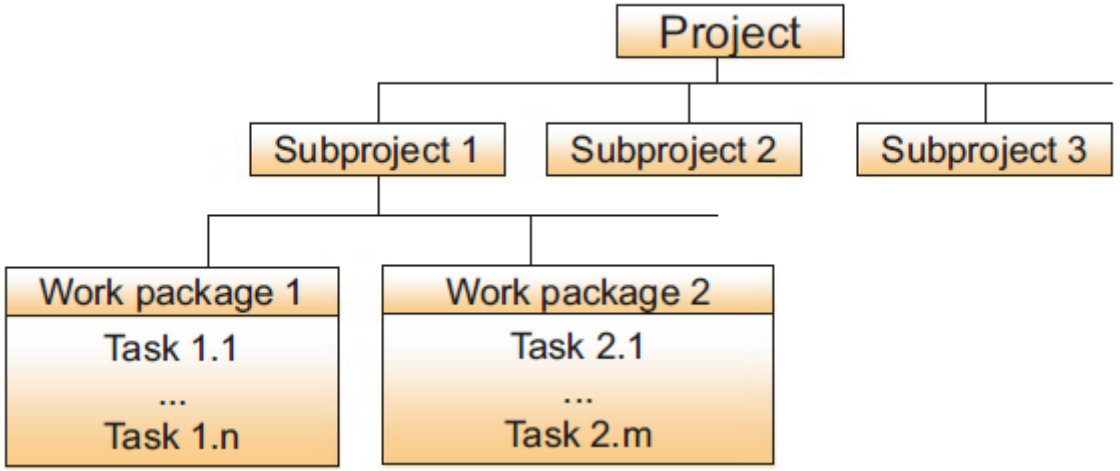
Planning a project: Overview of major acitivties



Thakkar, J.J. 2022. *Project Management: Strategic and Operational Planning*. Singapore: Springer.

Defining WHAT to do: Work Breakdown Structure (WBS)

- The WBS is the "complete hierarchical **representation of all elements (subprojects, work packages) of the project** structure as a diagram or list" to be executed by the project team **to accomplish the project objectives** and create the required deliverables.
- The hierarchical representation of the WBS aids in understanding the project deliverables and lists **all of them (focus: completeness!)**.
- WBS **interfaces with various project planning and scheduling activities** such as responsibility assignment matrix, **network scheduling, costing**, risk analysis, organizational structure, coordination of objectives, and control



Alam, D., and Gühl, U. 2022. *Project Management for Practice: A Guide and Toolbox for Successful Projects*. Berlin, Heidelberg: Springer.
Thakkar, J.J. 2022. *Project Management: Strategic and Operational Planning*. Singapore: Springer.

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Defining the HOW (long) and WHEN: Time Schedule

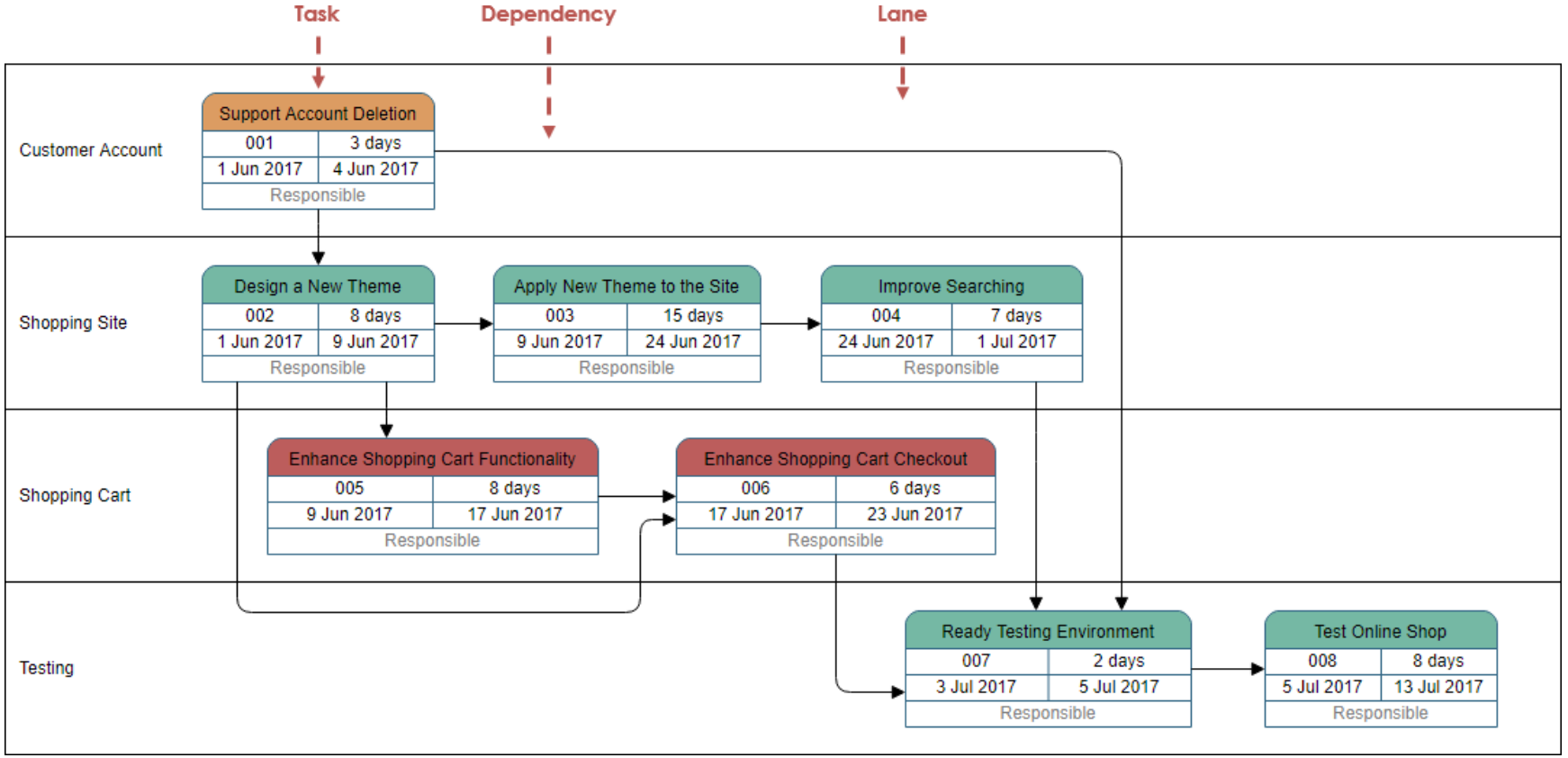
When the work involved in execution is catalogued in a WBS, the **duration of each task can now be estimated** and the **dependencies between tasks identified as follows:**

1. Estimate **duration**. The elapsed time (for example, in days) is estimated for each task in the WBS.
2. Define **dependencies among tasks**. For example, in an office move project, the task "Unload the vans" cannot start until "Drive vans to new location" is finished.
3. Define the **task schedule (start-/end dates)** by forward scheduling and backward scheduling
4. Identify the **critical path and slack**. The critical path is a chain of linked tasks that, if not completed on time, will necessarily delay the finish date of a project

Zwikael, O., and Smyrk, J.R. 2019. *Project Management: A Benefit Realisation Approach*, (1st ed.). Cham: Springer International Publishing AG.

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Defining the HOW (long) and WHEN: Network Charts provide a graphical view of a project's tasks and schedule

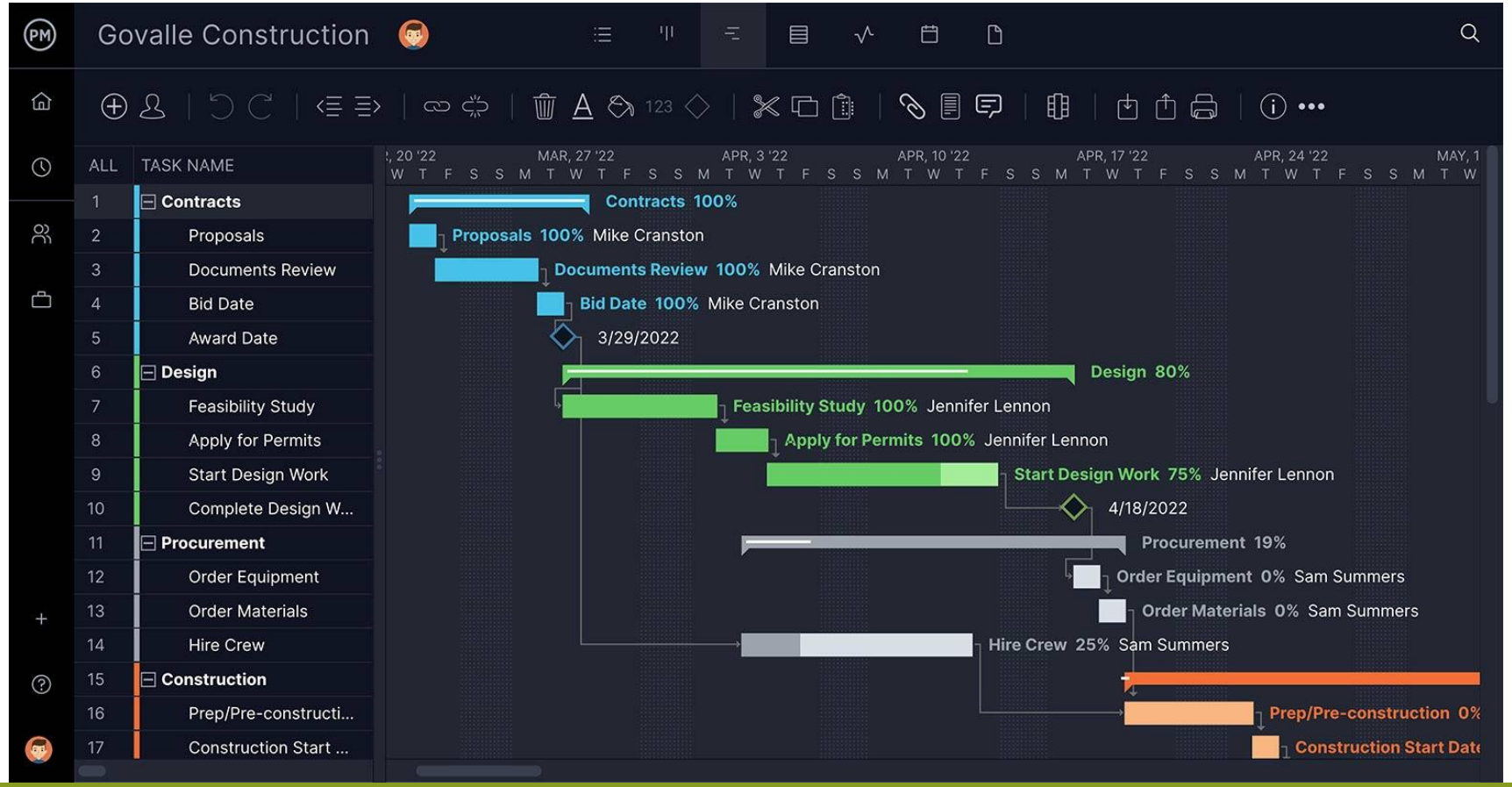


https://online.visual-paradigm.com/diagrams/tutorials/pert-chart-tutorial/

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An Alternative to Network Charts is the Gantt Chart

A Gantt chart is a bar chart that provides a **visual view of project tasks scheduled over time**.

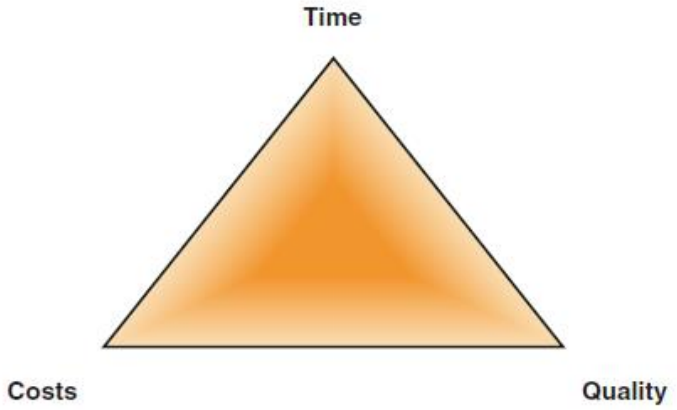


https://www.projectmanager.com/guides/gantt-chart

Milestone (Plan)



- According to DIN 69900, a milestone is an **“event of special importance”**.
- Milestones are a key component of project management, especially project controlling, and often define phase transitions.
- A milestone includes
  - a **date**,
  - **expected results** with verifiable criteria and
  - **estimation of budget/costs** needed to achieve the expected results



Alam, D., and Gühl, U. 2022. *Project Management for Practice: A Guide and Toolbox for Successful Projects*. Berlin, Heidelberg: Springer.

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Assignment

Create WBS and project schedule

Your local pizza dealer wants to start selling and delivering its products online through a website/e-commerce shop to increase its revenues.

1. Identify the essential activities needed to achieve the desired goals.
2. Structure the activities in the form of a WBS.
3. Now put all the tasks in a logical sequence and represent it graphically in the form of a network diagram or GANTT diagram.
4. Estimate the amount of work in PT (person-days) for the individual tasks.
5. Present your results to the class.

Project Management (PS)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

SAP UCC Magdeburg

Teaching material - Information

Teaching material - Version

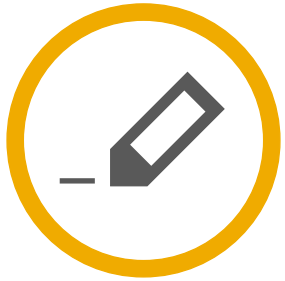
- 4.2 (August 2023)

- Software used
  - S/4HANA 2022
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



Authors

- Martin Wassmann
- Bret Wagner
- Stefan Weidner
- Babett Ruß



Target Audience

- Beginner

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Agenda

- PS Organizational Structure
- PS Master Data
- PS Processes
  - Project Planning
  - Project Execution
  - Project Closure

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Projects

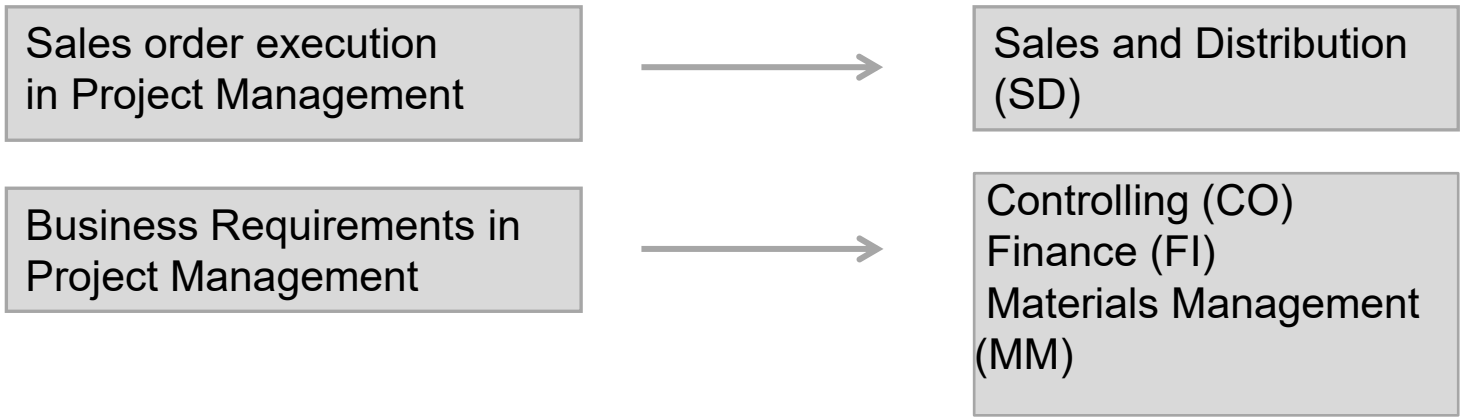
- Characteristics of projects:
  - Complex
  - Unique
  - High risk
  - Goal oriented
  - Limited in time, costly, high capacities needed
  - Strategic meaning
  - High quality standards
- Project types:
  - Customer projects
  - Overhead cost projects
  - Investment projects



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PS Organizational Structure (Project Management)

- Organizational structure needs to be defined in order to define execution of project
  - Project system itself does **not own its own organizational structure**
  - PS **inherits it's organization structures** via assignment into organizational units in the enterprise structure
  - The PS module is **integrated to the modules Finance and Logistics**
  - The productivity of this module is mainly created through the integration of other SAP modules



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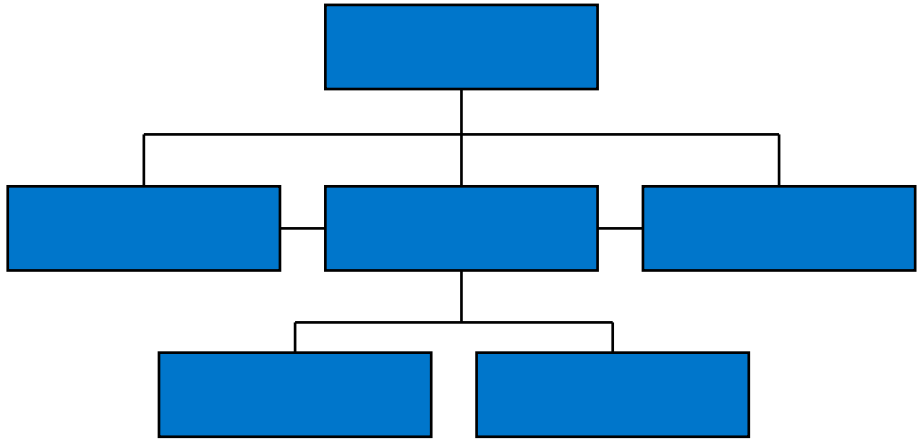
Agenda

- PS Organizational Structure
- PS Master Data
- PS Processes
  - Project Planning
  - Project Execution
  - Project Closure

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PS Master Data

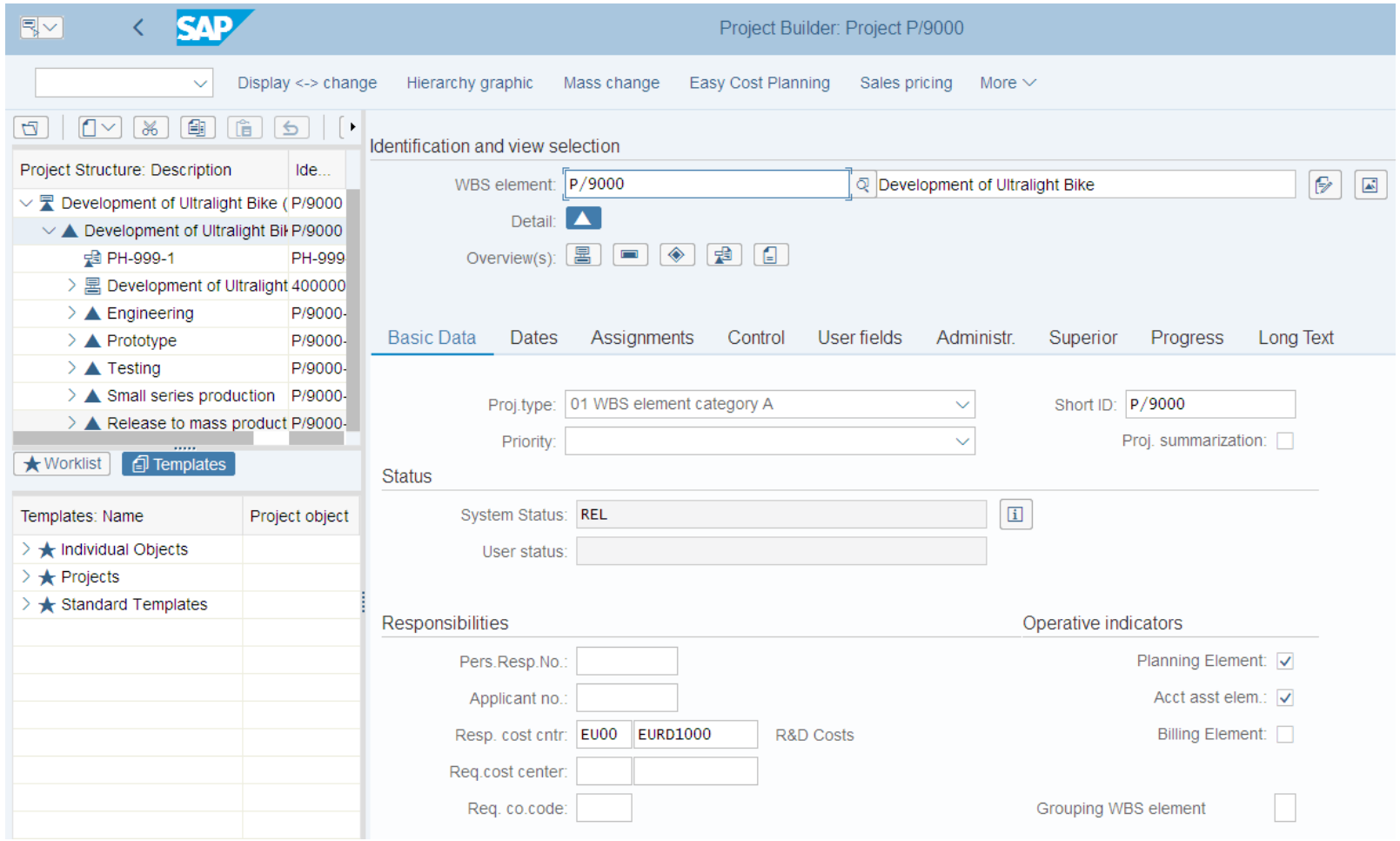
- Projects (Project Definition)
- WBS-Elements (Project Structure Plan Elements)
- Network (sequences of activities)
- Milestones
- Work Data



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Project Builder

- All data is prepared **using one transaction**
- Regularly performed projects can be re-used to the work of creating them.
- Scheduling of projects is based on networks

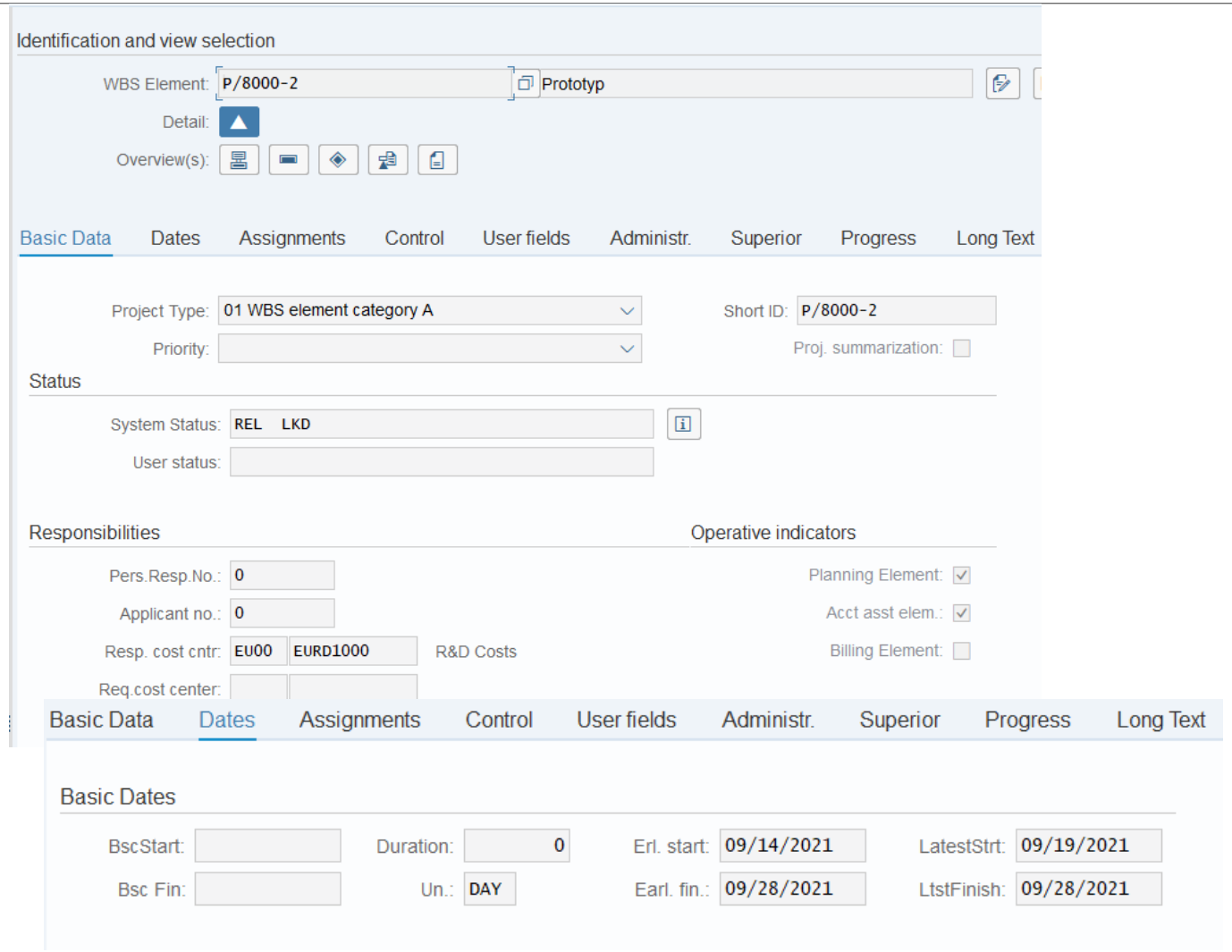


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WBS-Elements

- The WBS-Element describes either a **concrete task or a partial one** that can be further subdivided.

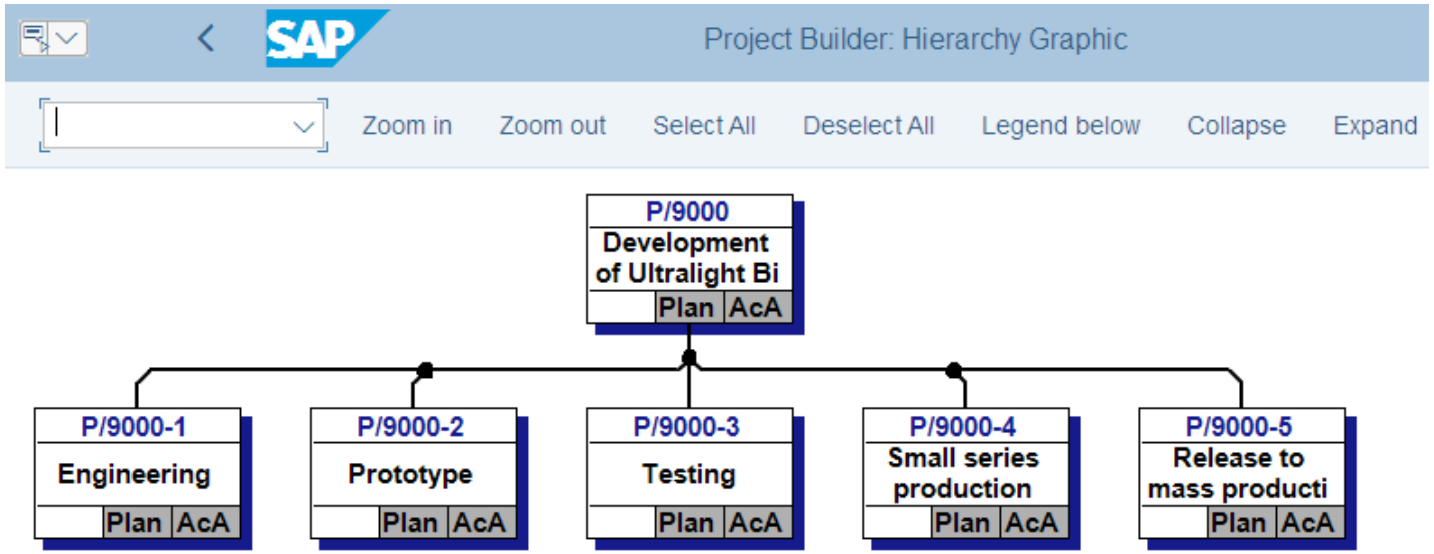
- Possible Status for WBS Elements
  - Released→ Entry of effective costs possible
  - Unlocked
  - Technically complete
  - Closed→ processes in Logistics and Finance
  - Deletion flag
  - Fully invoiced→ further invoices can be inhibited



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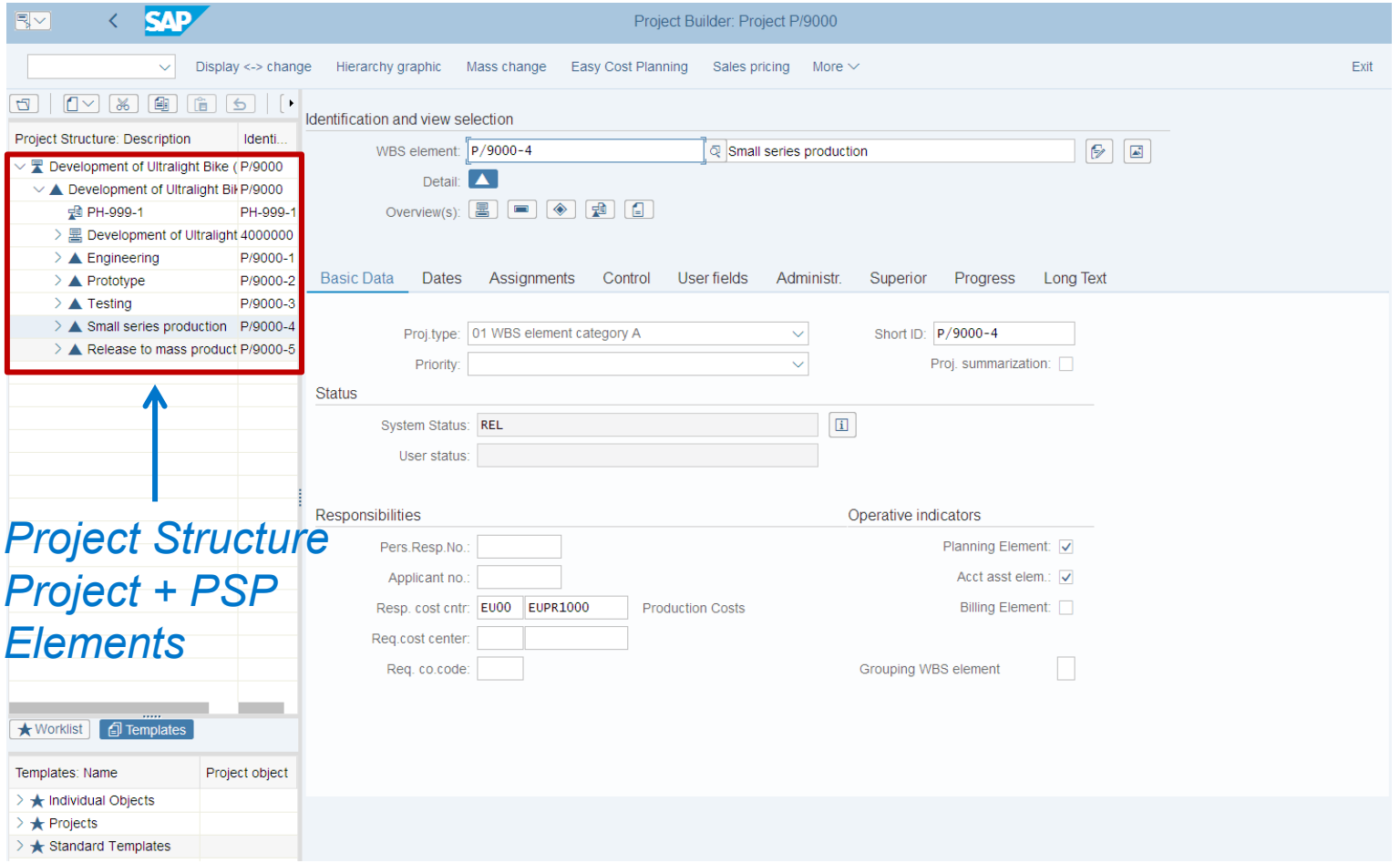
„Hierarchy Graphic“ visually represents a WBS in SAP

- The Hierarchy Graphic forms the basis for planning:
  - Costs
  - Revenues
  - Payments
  - Budgets
  - Schedules
- Can be structured in phases, functions and objects like conceptual design, overall planning or permissions.
- Project phases are, for instance:
  - Conceptual design
  - Overall planning
  - Permission, etc.



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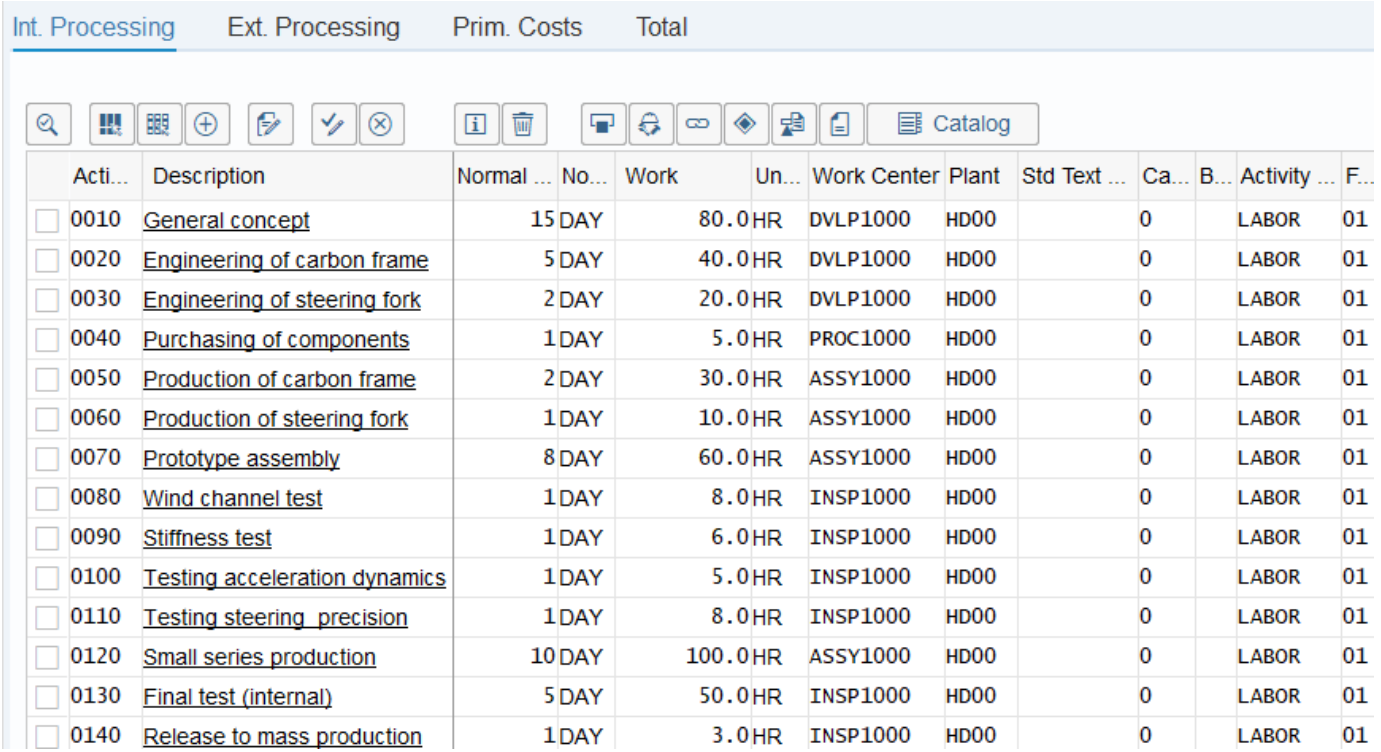
Project Structure in SAP S/4HANA



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Activities

- The Project System distinguishes between the following types of activity:
  - **Internally** processed activities
  - **Externally** processed activities
  - **General costs** activities
- The German standard DIN 69 900 describes activities as "components of a process " with the following characteristics:
  - They have a duration.
  - They have a defined start and a defined finish.
  - They are processed without interruption.
  - Resources are needed to execute them.
  - They incur costs.

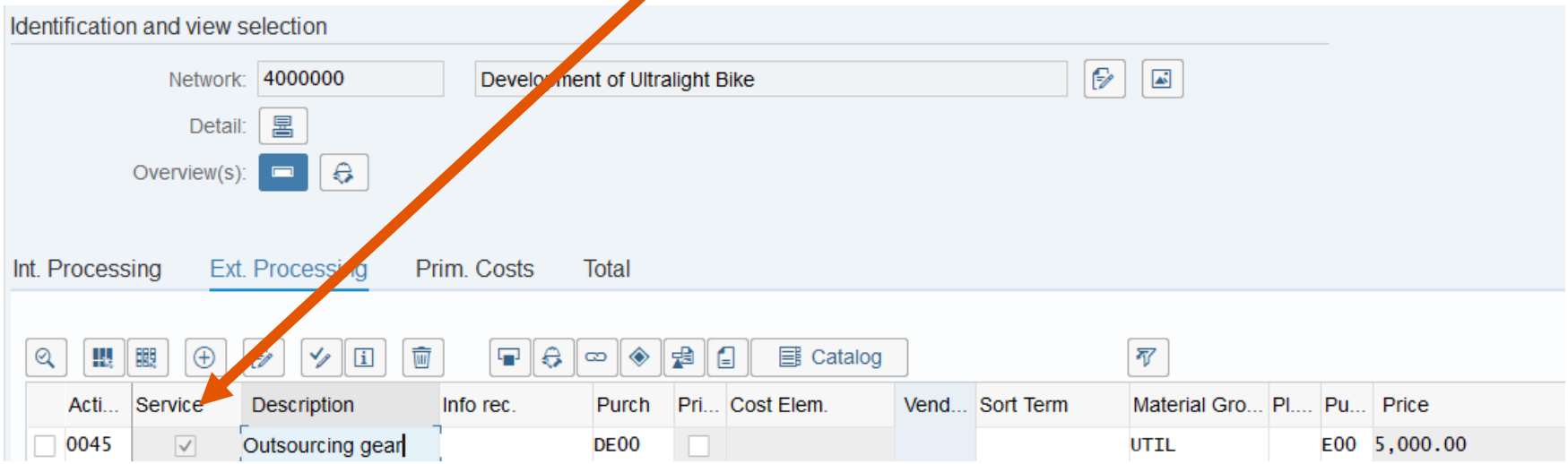


Acti...	Description	Normal ...	No...	Work	Un...	Work Center	Plant	Std Text ...	Ca...	B...	Activity ...	F...
0010	General concept	15	DAY	80.0	HR	DVLP1000	HD00		0		LABOR	01
0020	Engineering of carbon frame	5	DAY	40.0	HR	DVLP1000	HD00		0		LABOR	01
0030	Engineering of steering fork	2	DAY	20.0	HR	DVLP1000	HD00		0		LABOR	01
0040	Purchasing of components	1	DAY	5.0	HR	PROC1000	HD00		0		LABOR	01
0050	Production of carbon frame	2	DAY	30.0	HR	ASSY1000	HD00		0		LABOR	01
0060	Production of steering fork	1	DAY	10.0	HR	ASSY1000	HD00		0		LABOR	01
0070	Prototype assembly	8	DAY	60.0	HR	ASSY1000	HD00		0		LABOR	01
0080	Wind channel test	1	DAY	8.0	HR	INSP1000	HD00		0		LABOR	01
0090	Stiffness test	1	DAY	6.0	HR	INSP1000	HD00		0		LABOR	01
0100	Testing acceleration dynamics	1	DAY	5.0	HR	INSP1000	HD00		0		LABOR	01
0110	Testing steering_precision	1	DAY	8.0	HR	INSP1000	HD00		0		LABOR	01
0120	Small series production	10	DAY	100.0	HR	ASSY1000	HD00		0		LABOR	01
0130	Final test (internal)	5	DAY	50.0	HR	INSP1000	HD00		0		LABOR	01
0140	Release to mass production	1	DAY	3.0	HR	INSP1000	HD00		0		LABOR	01

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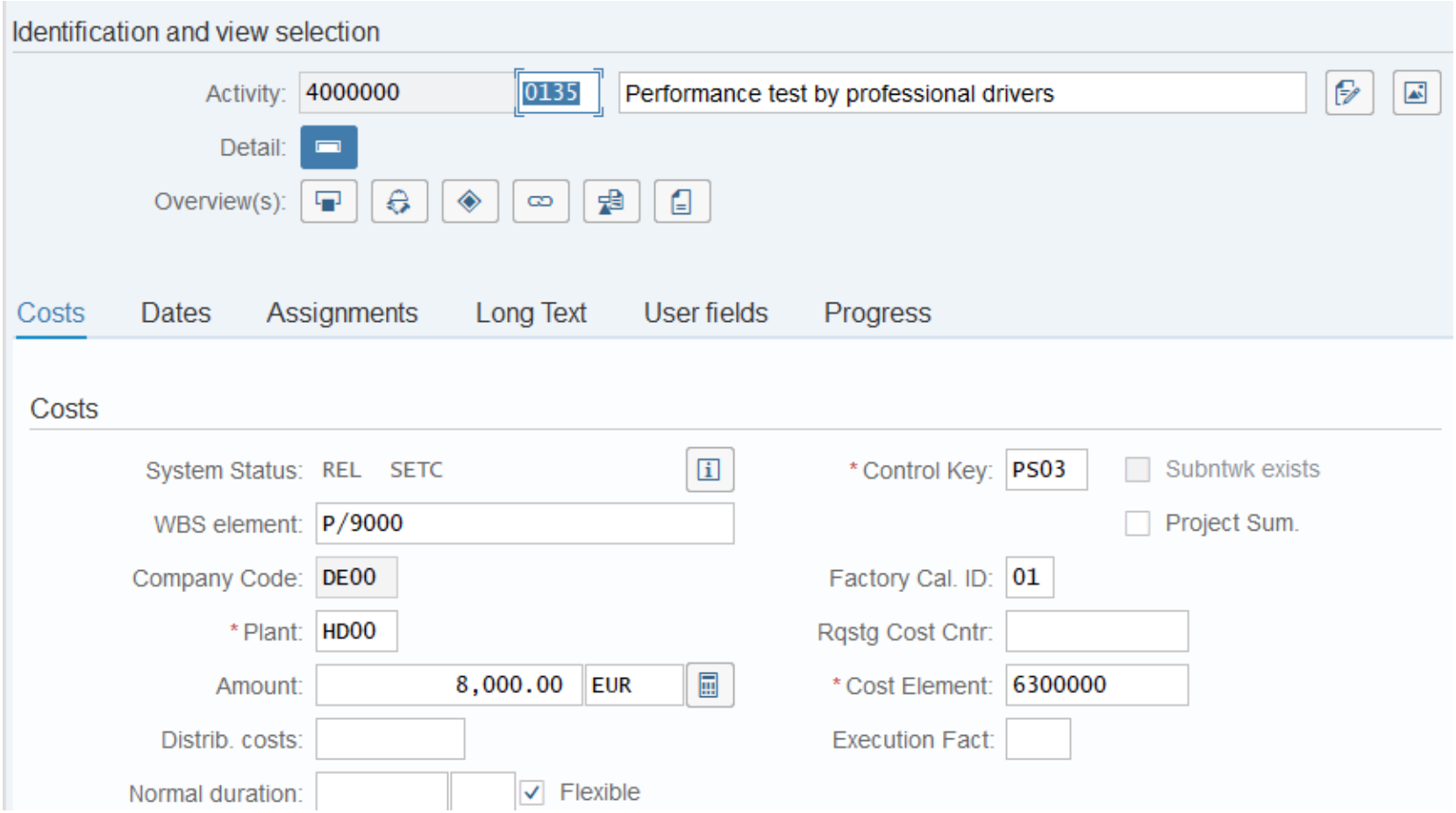
Externally processed activities/Services

- Planned
  - Related to activities that are collected in a service master
- Non-Planned
  - Related to activities that cannot be specified because it is not known, which individual specifications they comprise.
- For preparation of externally processed tasks, the corresponding control key must be entered in the network activity.



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Example: General costs activities

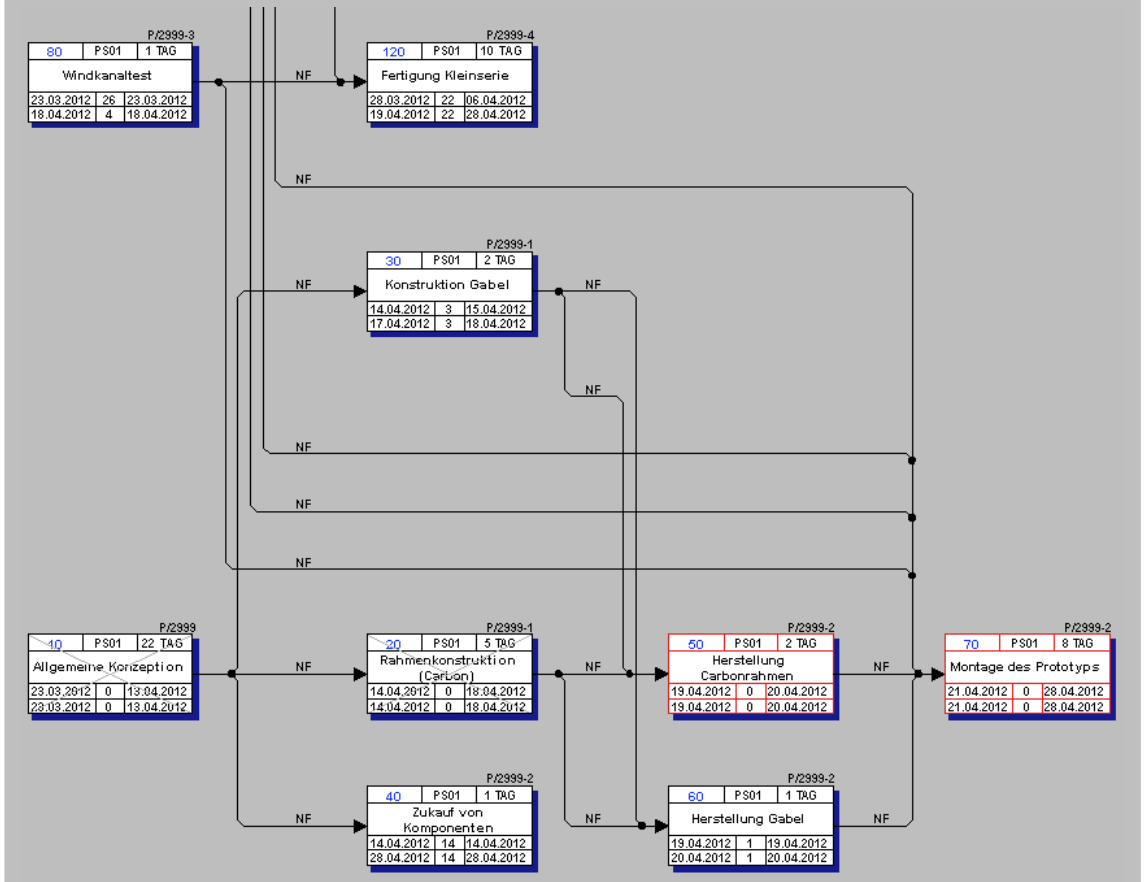


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Project Network Graph

- Depicts **activity sequences and relationships**
- Possibility to create activity sequences
- Change of internally processed activities to externally processed ones

Example Project Network Graph

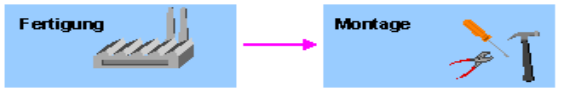


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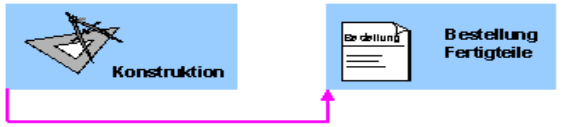
Project Network Graph: Relationships

- Relationships describe the **chronological sequence in a network plan oder standard network**. There can be dependencies of a technical or logical nature.

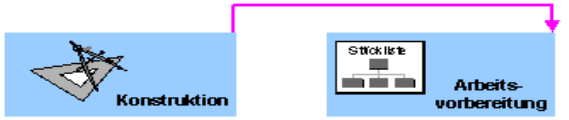
Finish to Start
A must finish before B can start.



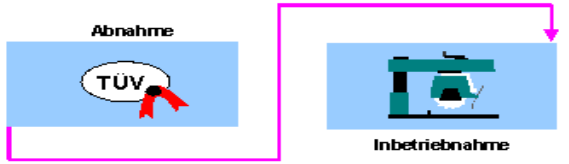
Start - Start
A must start before B can start.



Finish - Finish
A must finish before B can finish.



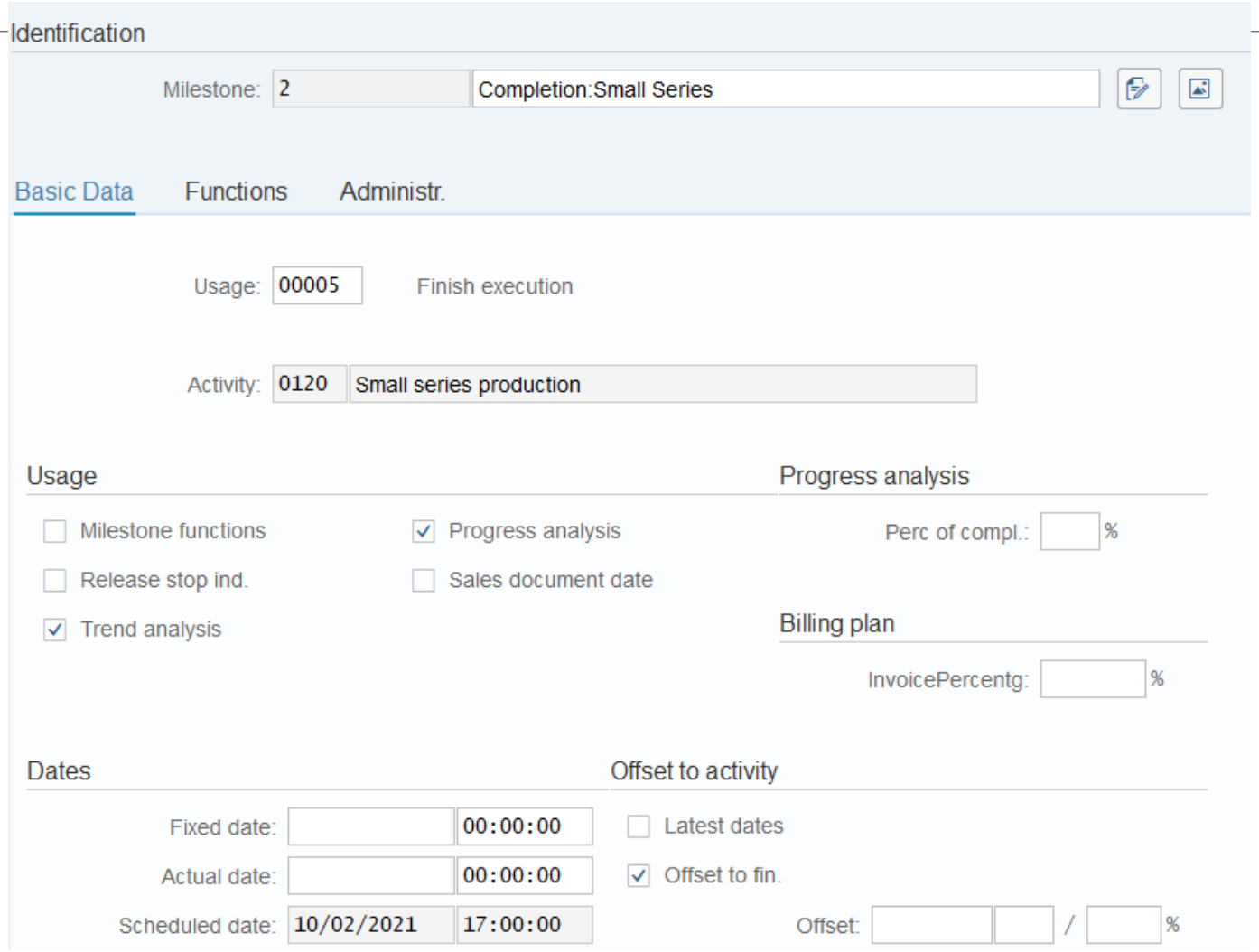
Start - Finish
A must start before B can finish.



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Milestones

- Milestones are used
  - to mark **transitions between different phases or departments**
  - to measure the progress of the project and mark, for example, decision points at which a committee must decide whether to continue the project or initiate the next phase.
- In the project system, they can be used for additional tasks and functions

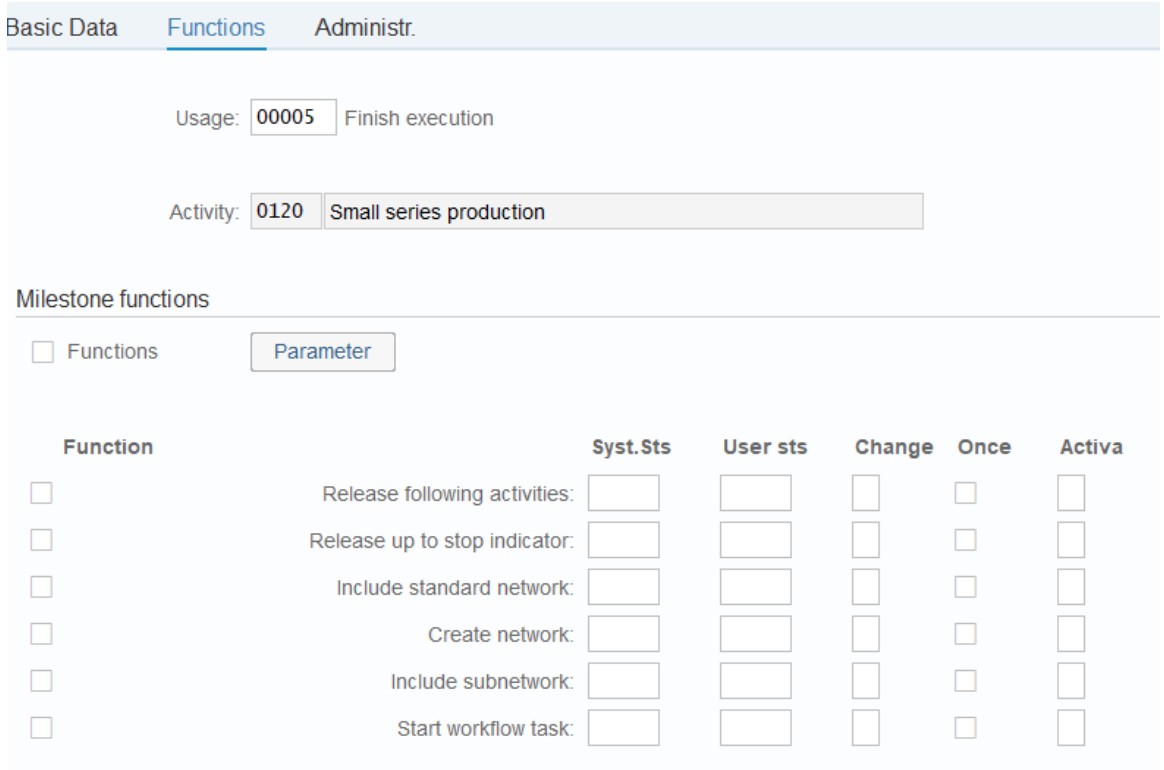


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Milestones

- Milestone functions **can be triggered when a system or application status changes** on the associated activity

- Functions:
  - Release of activities
  - Include standard schedule
  - Create hierarchy
  - Include parts of hierarchy
  - Start workflow tasks
  - etc.



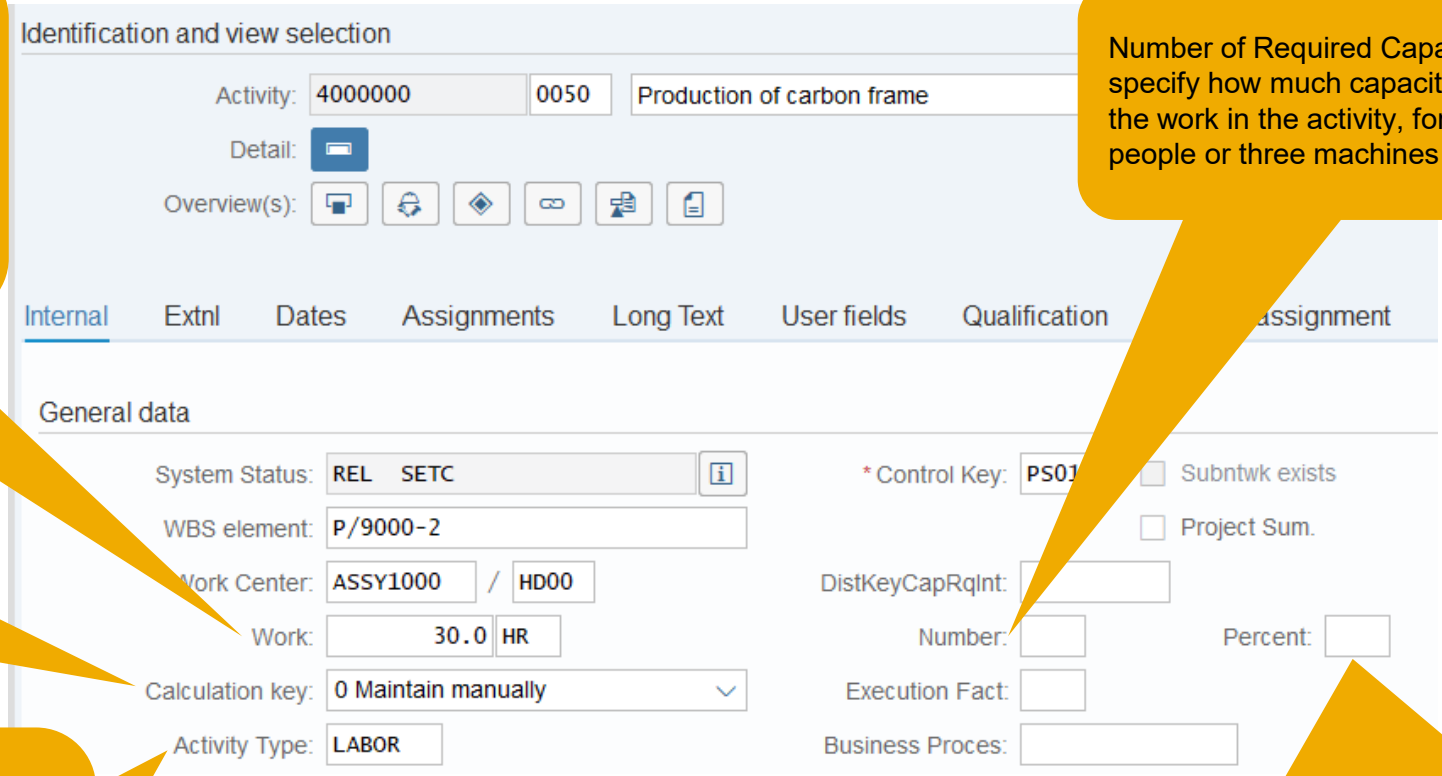
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Work Data in the Activity

Work: Here you specify the planned amount of work involved in carrying out the activity. For example, you can specify the engineering activity in hours or energy requirements in kilowatt hours.

Calculation key: You can maintain the data for work, duration, and number of capacities required **manually, or, you can let the system determine** it using the calculation key.

Activity type: If you want to cost networks, you must define an activity type for internal processing and a formula key for calculating costs. When you enter an activity type you can display different costs for special qualifications or weekend work.



Number of Required Capacities: Here you specify how much capacity is required to do the work in the activity, for example, two people or three machines

Percent: With this value you specify the percentage of the work that is used in the activity. The percentage is only used when you let the system calculate duration, work, or the number of capacities using the calculation key. If you do not enter a percentage, the system will assume 100%.

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Agenda

- PS Organizational Structure
- PS Master Data
- PS Processes
  - Project Planning
  - Project Execution
  - Project Closure

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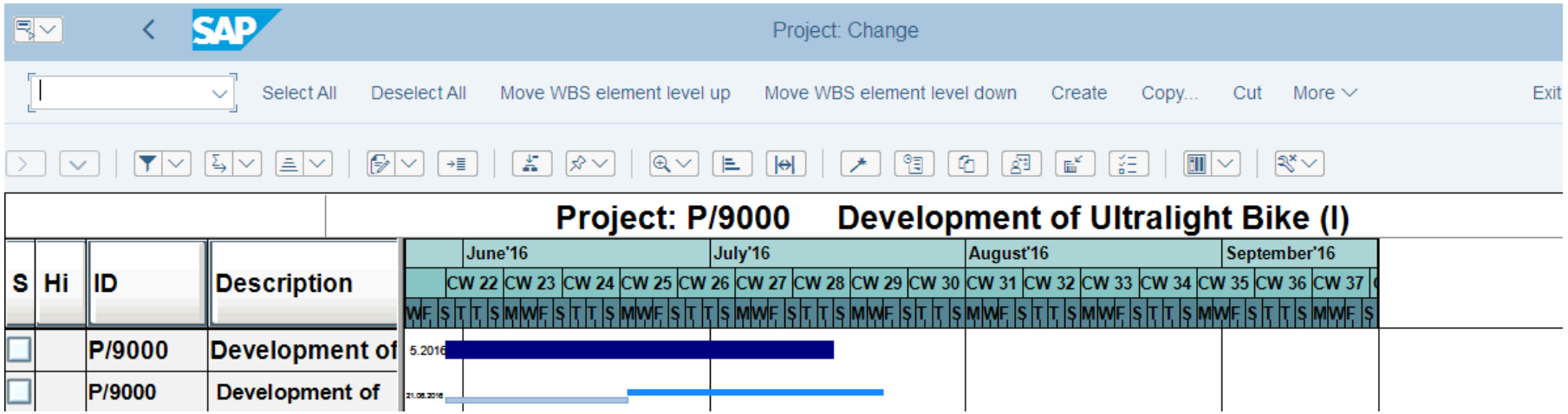
PS Processes

- Project Planning
- Project Execution

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Project Planning – Project Planning Board

- Graphical interface for creation and preparation of projects
- Function:
  - Termination (of resources)
  - Coordinate work
  - Check capacities
  - Calculate costs



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Project Execution

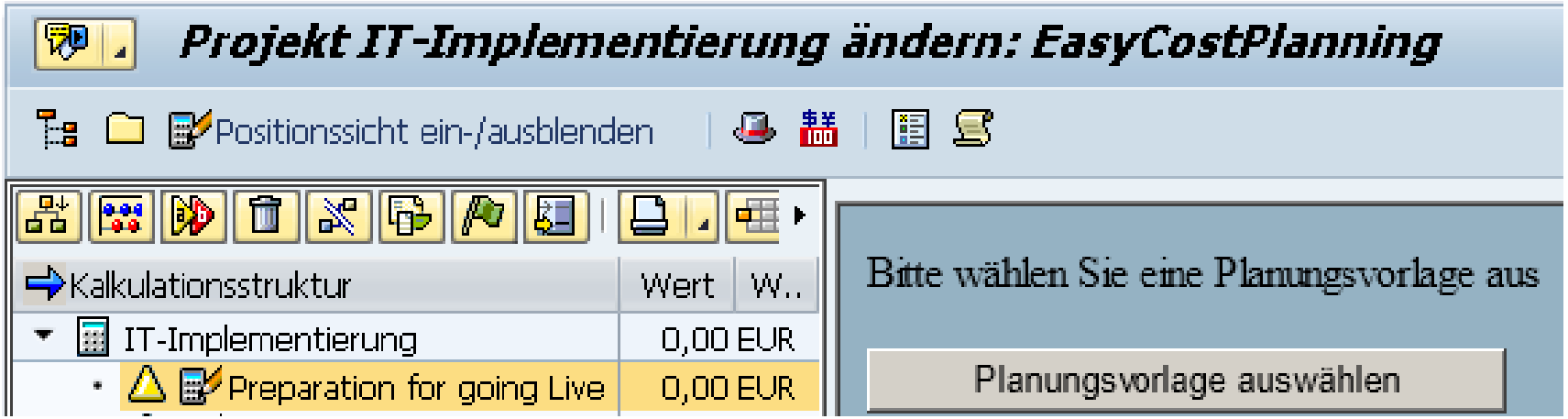
- Project execution comprises several steps (e.g. purchase process)
- **Purchase requests** are generated automatically using the hierarchy
- When **goods receipt of services** is executed, services and their approval are recorded
- With the execution of **services, liabilities and costs for items** that were planned using Easy Cost Planning are entered

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Project Execution: Easy Cost Planning

Easy Cost Planning enables cost planning based on amounts and characteristics. It is executed on the level of project structures. Main advantages of these is a fast, easy and clearly arranged data acquisition.

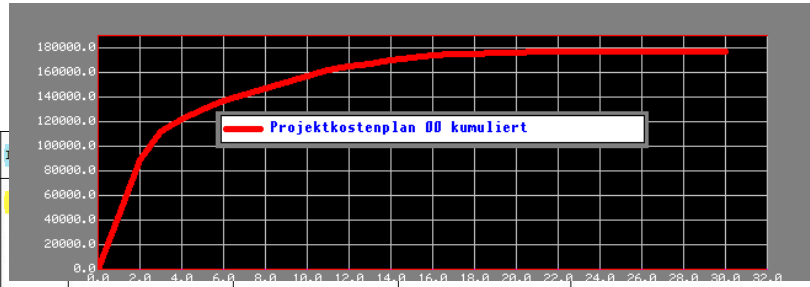
- Possible entries with Easy Cost Planning:
  - Internal accounting for services
  - Direct process accounting
  - Reservations
  - Goods issue
  - Purchase requisition
  - Sales order



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Information System

- The information system is used to evaluate **key performance indicators** within a project.
- Provides the user with **available reports and graphs**
- Overview over **performance, schedule and costs**
- Report types:
  - Structure-based
  - Cost-oriented
  - Item reports
  - Summarised reports



Projektelement	Objekttyp	FrStrt E	SpEnde E	Ist-Star				
Entwicklung Ultraleicht-Rad 99	Projektdef	21.02.21	07.04.21					
Entwicklung Ultraleichtrad	PSP-Elemen	21.02.21	07.04.21					
PH-999-1	PS-Text							
Entwicklung Ultraleichtrad	Netzplan	21.02.21	07.04.21					
Allgemeine Konzeption	Eigenbearb	21.02.21	07.03.21		10 450,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Fremdvergabe Schaltung	Fremdbearb	07.03.21	14.03.21		5 000,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
PH-003-1	PS-Text							
Performance-Test durch P	Kostenvorg	03.04.21	06.04.21		8 000,00 EUR	9 700,00 EUR	0,00 EUR	0,00 EUR
Konstruktion	PSP-Elemen	08.03.21	13.03.21		6 600,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Temp	Meilenstei							
Rahmenkonstruktion (Carb	Eigenbearb	08.03.21	12.03.21		4 400,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Konstruktion Gabel	Eigenbearb	08.03.21	13.03.21		2 200,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Prototyp	PSP-Elemen	08.03.21	22.03.21		5 250,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Zukauf von Komponenten	Eigenbearb	08.03.21	14.03.21		250,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Herstellung Carbonrahmen	Eigenbearb	13.03.21	14.03.21		1 500,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Herstellung Gabel	Eigenbearb	10.03.21	14.03.21		500,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Montage des Prototyps	Eigenbearb	15.03.21	22.03.21		3 000,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Fertigstellung: Protot	Meilenstei	22.03.21	22.03.21					
Testverfahren	PSP-Elemen	23.03.21	23.03.21		1 350,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Windkanaltest	Eigenbearb	23.03.21	23.03.21		400,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Steifigkeitstest	Eigenbearb	23.03.21	23.03.21		300,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Test Beschleunigungsdyna	Eigenbearb	23.03.21	23.03.21		250,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Test Lenkpräzision	Eigenbearb	23.03.21	23.03.21		400,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Kleinserie	PSP-Elemen	24.03.21	02.04.21		5 000,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Fertigung Kleinserie	Eigenbearb	24.03.21	02.04.21		5 000,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Abschluss: Kleinserie	Meilenstei	02.04.21	02.04.21					
Freigabe an Produktion	PSP-Elemen	03.04.21	07.04.21		2 650,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Abschlusstest (intern)	Eigenbearb	03.04.21	07.04.21		2 500,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Freigabe an Produktion	Eigenbearb	07.04.21	07.04.21		150,00 EUR	0,00 EUR	0,00 EUR	0,00 EUR
Abschluss: Freigabe	Meilenstei	07.04.21	07.04.21					

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Period Closing

- In general, one time per period executed in background
- To execute billing, rules are necessary
- Costs result from vendor invoices, customer billing, etc. ➔ they are transferred to FI and other modules.

Direct settlement
Each object is settled directly to the external cost object (examples: cost center, profitability segment)

Multi-level settlement
You settle individual objects in the project to the higher-level WBS element first. This then settles the costs collected to the external receiver.

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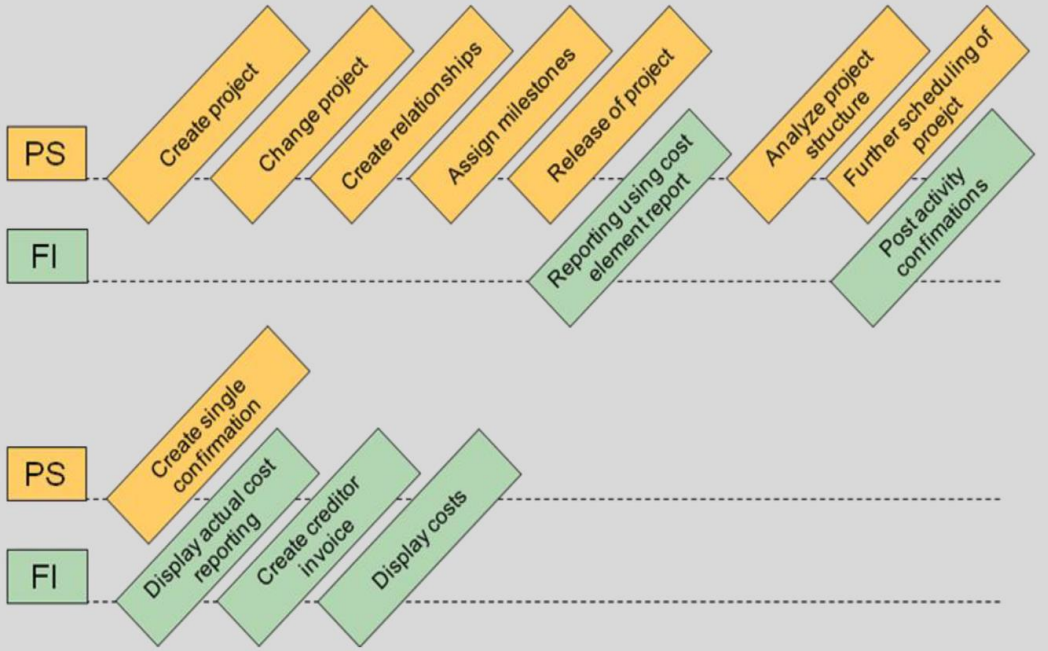
Agenda

- General Foundations Project Management
- PS Organizational Structure
- PS Master Data
- PS Processes
  - Project Planning
  - Project Execution
  - Project Closure
- Outlook PS Case Study

Case Study PS

At the beginning of your project you assign WBS elements, activities and milestones at Project Builder. Also, you assign milestones to your project.

In the last few steps, you create a creditor invoice for your business partner, who acts as an intermediary for the test driver and you have to pay for the entire project.



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Learning Objectives (What?)

Students can…

- explain why organizations employ projects to achieve their objectives.
- explain the relationships between portfolios, programmes and projects.
- name important processes in Project Management.
- describe a generalized project planning approach including major planning artefacts.
- explain the concepts of a work breakdown structure (WBS) and network chart.
- explain relevant master data important for the PS module.